

$$1) \quad \omega_e = \frac{2\pi \text{ rad}}{24 \times 3600 \text{ s}} = 7.27 \times 10^{-5} \frac{\text{rad}}{\text{s}} \quad , \quad R_e = 6.37 \times 10^6 \text{ m}$$

$$\omega_e^2 R_e = 3.37 \times 10^{-2} \frac{\text{m}}{\text{s}^2}$$

$$\omega_e R_e = 463.1 \frac{\text{m}}{\text{s}} \Rightarrow \text{Earth}$$

$$\omega_v = \frac{2\pi \text{ rad}}{243 \times 24 \times 3600 \text{ s}} = 3.0 \times 10^{-7} \frac{\text{rad}}{\text{s}} \quad , \quad R_v = 6.05 \times 10^6 \text{ m}$$

it takes 243
days for Venus
to rotate once
about its axis

$$\omega_v^2 R_v = 5.42 \times 10^{-7} \frac{\text{m}}{\text{s}^2}$$

$$\omega_v R_v = 1.806 \frac{\text{m}}{\text{s}} \Rightarrow \text{Venus}$$

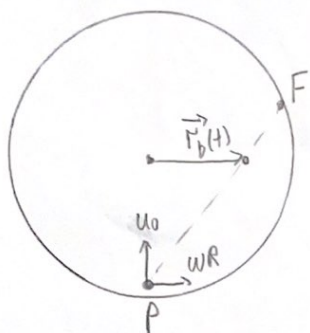
$$\omega_J = \frac{2\pi \text{ rad}}{10 \times 3600 \text{ s}} = 1.74 \times 10^{-4} \frac{\text{rad}}{\text{s}} \quad , \quad R_J = 6.99 \times 10^7 \text{ m}$$

1 rotation
takes 10 hours

$$\omega_J^2 R_J = 2.11 \frac{\text{m}}{\text{s}^2}$$

$$\omega_J R_J = 12162 \frac{\text{m}}{\text{s}} \Rightarrow \text{Jupiter}$$

2)
a)



→ Initial displacement of the ball:

$$x(0) = 0, y(0) = 0 \Rightarrow \vec{r}_b(0) = -R\hat{j}$$

→ Initial velocity of the ball:

$$\dot{x}(0) = |\vec{\omega} \times \vec{R}| = \omega R \rightarrow \text{since } \vec{\omega} \text{ is in } \hat{k} \text{ direction}$$

$$\dot{y}(0) = u_0$$

$$\vec{v}_b(0) = \omega R \hat{i} + u_0 \hat{j}$$

→ Here we have utilized: $\left(\frac{d\vec{r}_b}{dt}\right)_{in} = \left(\frac{d\vec{r}_b}{dt}\right)_{rot} + \vec{\Omega} \times \vec{r}_b$

$$\left(\frac{d\vec{r}_b}{dt}\right)_{rot} = u_0 \hat{j} \Rightarrow \text{ball's velocity w.r.t. lab frame at } t=0$$

$$\vec{\Omega} \times \vec{r}_b = \omega \hat{k} \times R(-\hat{j}) = \omega R \hat{i}$$

→ In the inertial reference frame ball is not subject to any force since the lab is rotating in empty space. That means the ball will follow a linear trajectory in the inertial reference frame. W

$$\left. \begin{aligned} x(t) &= x(0) + \dot{x}(0)t = \omega R t \\ y(t) &= y(0) + \dot{y}(0)t = -R + u_0 t \end{aligned} \right\} \boxed{\vec{r}_b(t) = \omega R t \hat{i} + (u_0 t - R) \hat{j}}$$

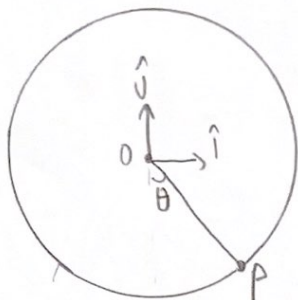
→ Ball will hit to side of the lab when:

$$x^2(t) + y^2(t) = R^2$$

$$\omega^2 R^2 t^2 + R^2 + u_0^2 t^2 - 2u_0 R t = R^2 ; (\omega^2 R^2 + u_0^2) t'^2 = 2u_0 R t'$$

$$\boxed{t' = \frac{2u_0 R}{\omega^2 R^2 + u_0^2}}$$

b)



$$\vec{r}_p(t) = -R \cos \theta \hat{j} + R \sin \theta \hat{i}$$

$$\vec{r}_p(t) = +R \sin(\omega t) \hat{i} - R \cos(\omega t) \hat{j}$$

* After an arbitrary time t s.t. $\omega t = \theta$ has passed.

$$c) \vec{p} = \vec{r}_b - \vec{r}_p$$

$$\vec{p}(t) = R(\omega t - \sin \omega t) \hat{i} + [u_0 t + R(\cos \omega t - 1)] \hat{j}$$

Let's investigate the distance between point P and the ball when it hits the lab:

$$\vec{p}(t') = \vec{p}\left(t = \frac{2u_0 R}{\omega^2 R^2 + u_0^2}\right) = R \left[\frac{2u_0 \omega R}{\omega^2 R^2 + u_0^2} - \sin\left(\frac{2u_0 \omega R}{\omega^2 R^2 + u_0^2}\right) \right] \hat{i}$$

$$+ R \left[\frac{2u_0^2}{\omega^2 R^2 + u_0^2} + \cos\left(\frac{2u_0 \omega R}{\omega^2 R^2 + u_0^2}\right) - 1 \right] \hat{j}$$

Letting $\sigma = \frac{u_0}{\omega R}$

$$\hookrightarrow \vec{p}(t') = R \left[\frac{2\sigma}{1+\sigma^2} - \sin\left(\frac{2\sigma}{1+\sigma^2}\right) \right] \hat{i} + R \left[\frac{2\sigma^2}{1+\sigma^2} - 1 + \cos\left(\frac{2\sigma}{1+\sigma^2}\right) \right] \hat{j}$$

If we Taylor expand the sin and cos terms using computation programs:

$$\sin\left(\frac{2\sigma}{1+\sigma^2}\right) = 2\sigma - \frac{10\sigma^3}{3} + \frac{94\sigma^5}{15} + \dots \text{ (higher order terms)}$$

$$\cos\left(\frac{2\sigma}{1+\sigma^2}\right) = 1 - 2\sigma^2 + \frac{14\sigma^4}{3} + \dots$$

$$\vec{p}(t') = R \left[\frac{2\sigma}{1+\sigma^2} - \left(2\sigma - \frac{10\sigma^3}{3} + \frac{94\sigma^5}{15} + \dots \right) \right] \hat{i} \\ + R \left[\frac{2\sigma^2}{1+\sigma^2} - 1 + \left(1 - 2\sigma^2 + \frac{14\sigma^4}{3} + \dots \right) \right] \hat{j}$$

$$\vec{p}(t') = R \left[\frac{2\sigma}{1+\sigma^2} + \frac{-2\sigma - \sigma^3}{1+\sigma^2} + \frac{10\sigma^3}{3} - \frac{94\sigma^5}{15} + \dots \right] \hat{i} \\ + R \left[\frac{2\sigma^2}{1+\sigma^2} + \frac{-2\sigma^2 - 2\sigma^4}{1+\sigma^2} + \frac{14\sigma^4}{3} + \dots \right] \hat{j}$$

$$= R \left[\frac{-\sigma^3}{1+\sigma^2} + \frac{10\sigma^3}{3} - \frac{94\sigma^5}{15} \right] \hat{i} + R \left[\frac{-2\sigma^4}{1+\sigma^2} + \frac{14\sigma^4}{3} + \dots \right] \hat{j}$$

⇒ What we can infer from the calculations are the following:

→ When σ is really small ($\sigma \ll 1$), which means u_0 is very small compared to ωR ; 3rd and higher order terms can be neglected. So $\vec{p}(t')$ becomes:

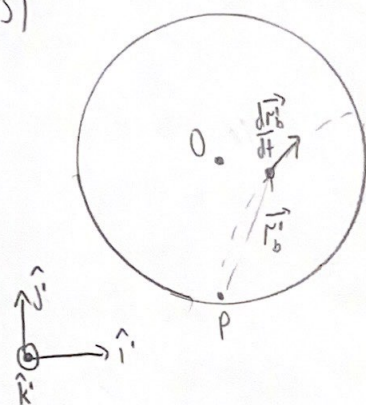
$$\lim_{\sigma \rightarrow 0} \vec{p}(t') = 0$$

↳ If we throw the ball with a small enough upward speed, it will almost fall into the same position. This can be observed in Earth. Because ωR term dominates u_0 , when we throw ^{upward} a ball into the air its deflection is negligible and it falls back to the same spot.

→ However u_0 becomes comparable to ωR in the motion of high speed projectiles or rockets. Certain deflection will be observed.

→ When σ is very large ($\sigma \gg 1$) which means u_0 is much bigger than ωR ; we observe that $\vec{p}(t') = 2R\hat{j}$ which makes sense. Ball is thrown so fast that it hits the opposite side of the lab before P can barely rotate.

3)



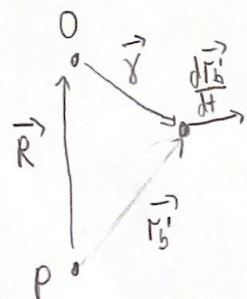
* In a rotating reference frame equations of motion turns out to be (assuming $\vec{\omega}$ is constant)

$$m\vec{a}_{rot} = \vec{F} - m2\vec{\omega} \times \vec{v}_{rot} - m\vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$\vec{F} = 0$ since no forces acting on the ball in the inertial frame

→ Here $\vec{r}$ must be the ^{position} vector from the axis of rotation, to the objects location (shown in question 7)

→ Define $\vec{\gamma} = \vec{r}_b - \vec{R}$



$$\frac{d^2 \vec{r}_b}{dt^2} = -2\vec{\omega} \times \frac{d\vec{r}_b}{dt} - \vec{\omega} \times (\vec{\omega} \times \vec{\gamma})$$

→ Separate $\vec{r}_b$ into $\hat{i}$ and $\hat{j}$ components: $\vec{r}_b(t) = x\hat{i} + y\hat{j}$

$$\frac{d^2 x}{dt^2} \hat{i} + \frac{d^2 y}{dt^2} \hat{j} = -2\omega \hat{k} \times \left(\frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} \right) - \vec{\omega} \times \left\{ \vec{\omega} \times [x\hat{i} + (y-R)\hat{j}] \right\}$$

$$= -2\omega \frac{dx}{dt} \hat{j} + 2\omega \frac{dy}{dt} \hat{i} - \vec{\omega} \times (\omega x \hat{j} - \omega(y-R)\hat{i})$$

$$= -2\omega \frac{dx}{dt} \hat{j} + 2\omega \frac{dy}{dt} \hat{i} + \omega^2 x \hat{i} + \omega^2 (y-R) \hat{j}$$

$$\frac{d^2 x}{dt^2} = 2\omega \frac{dy}{dt} + \omega^2 x$$

$$\frac{d^2 y}{dt^2} = -2\omega \frac{dx}{dt} + \omega^2 (y-R)$$

with initial conditions
 $x(0)=0, y(0)=0, \frac{dx}{dt}(0)=0, \frac{dy}{dt}(0)=u_0$
 constitutes our system of linear differential equations that defines the ball's motion

→ In the previous question by Taylor expanding the expression for $\vec{p}(t')$ we have seen that x component goes to zero as $\sigma \rightarrow 0$. So when $u_0 \ll \omega R$ we can make the approximation that $x \approx 0$. This is not true for y because y component changes during the motion of the ball however x component can only grow in one direction. Hence if $x \approx 0$ when it hits the lab, $x \approx 0$ during the entire motion.

→ By plotting the exact expressions for $\dot{x}$ and $\ddot{x}$, I have observed they don't stay small and bounded near 0 so we can't neglect them

→ With the approximation $x \approx 0$, our system of D.E. turns into.

$$(1) \quad \ddot{x} = 2\omega \dot{y} \quad \text{with } x(0) = y(0) = 0, \quad \dot{x}(0) = 0, \quad \dot{y}(0) = u_0$$

$$(2) \quad \ddot{y} = -2\omega \dot{x} + \omega^2(y - R)$$

$$(1) \Rightarrow \dot{x} - \dot{x}(0) = 2\omega [y - y(0)] \Rightarrow \dot{x} = 2\omega y$$

$$(2) \Rightarrow \ddot{y} = -2\omega(2\omega y) + \omega^2(y - R) = -4\omega^2 y + \omega^2 y - \omega^2 R$$

$$\ddot{y} + 3\omega^2 y = -\omega^2 R \Rightarrow y = A \sin(\sqrt{3}\omega t) + B \cos(\sqrt{3}\omega t) - \frac{R}{3}$$

$$y(0) = B - \frac{R}{3} = 0 \Rightarrow B = \frac{R}{3}, \quad \dot{y}(0) = A\sqrt{3}\omega = u_0 \Rightarrow A = \frac{u_0}{\sqrt{3}\omega}$$

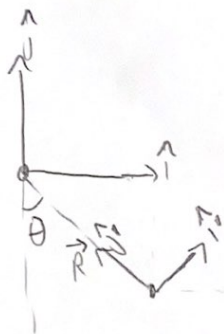
$$y(t) = \frac{u_0}{\sqrt{3}\omega} \sin(\sqrt{3}\omega t) + \frac{R}{3} [\cos(\sqrt{3}\omega t) - 1]$$

$$x(t) = 2\omega \int_0^t y(t) dt = 2\omega \left[\frac{u_0}{3\omega^2} [\cos(\sqrt{3}\omega t) - 1] - \frac{R}{3} \left[\frac{\sin(\sqrt{3}\omega t)}{\sqrt{3}\omega} - t \right] \right]$$

$$x(t) = \frac{R}{3} \left[\frac{\sin(\sqrt{3}\omega t)}{\sqrt{3}\omega} - t \right] - \frac{2u_0}{3\omega} \cos(\sqrt{3}\omega t)$$

↳ These solutions approximately give the "trajectory of the ball

4)



$$\hat{i}' = \hat{i} \cos \theta + \hat{j} \sin \theta$$

$$\hat{j}' = -\hat{i} \sin \theta + \hat{j} \cos \theta$$

$$\Leftrightarrow \begin{cases} \hat{i} = \hat{i}' \cos \theta - \hat{j}' \sin \theta \\ \hat{j} = \hat{i}' \sin \theta + \hat{j}' \cos \theta \end{cases}$$

* Since 2 reference frames are not co-centric

$$\vec{V}_i' = O_{ia} V_a + R \hat{j}'$$

$$\vec{r}_b(t) = \omega R t \hat{i} + (u_0 t - R) \hat{j} \Rightarrow \text{in inertial frame}$$

$$\begin{aligned} \vec{r}_b'(t) &= [\omega R t \cos \theta + (u_0 t - R) \sin \theta] \hat{i}' \\ &\quad + [-\omega R t \sin \theta + (u_0 t - R) \cos \theta + R] \hat{j}' \end{aligned} \Rightarrow \text{in rotating frame}$$

$$\begin{aligned} \vec{r}_b'(t) &= [R \omega t \cos(\omega t) + (u_0 t - R) \sin(\omega t)] \hat{i}' \\ &\quad + [-R \omega t \sin(\omega t) + (u_0 t - R) \cos(\omega t) + R] \hat{j}' \end{aligned}$$

* Since P is stationary and at the origin w.r.t. rotating frame:

$$\vec{r}_P' = 0$$

$$\begin{aligned} \vec{p}'(t) = \vec{r}_b'(t) &= [R \omega t \cos(\omega t) + (u_0 t - R) \sin(\omega t)] \hat{i}' \\ &\quad + [-R \omega t \sin(\omega t) + (u_0 t - R) \cos(\omega t) + R] \hat{j}' \end{aligned} //$$

5)

For any vector $\vec{V}$ we have found that:

$$\left(\frac{d\vec{V}}{dt}\right)_{in} = \left(\frac{d\vec{V}}{dt}\right)_{rot} + \vec{\omega} \times \vec{V}$$

$\nearrow \vec{V}_{in}$ $\nearrow \vec{V}_{rot}$

For the position vector: $\left(\frac{d\vec{r}}{dt}\right)_{in} = \left(\frac{d\vec{r}}{dt}\right)_{rot} + \vec{\omega} \times \vec{r}$

$$\vec{a}_{in} = \left(\frac{d\vec{V}_{in}}{dt}\right)_{in} = \left(\frac{d\vec{V}_{in}}{dt}\right)_{rot} + \vec{\omega} \times \vec{V}_{in}$$

$$= \frac{d}{dt} \left(\vec{V}_{rot} + \vec{\omega} \times \vec{r} \right)_{rot} + \vec{\omega} \times \left(\vec{V}_{rot} + \vec{\omega} \times \vec{r} \right)$$

$$= \frac{d\vec{V}_{rot}}{dt} + \left(\frac{d\vec{\omega}}{dt}\right)_{rot} \times \vec{r} + \vec{\omega} \times \vec{V}_{rot} + \vec{\omega} \times \vec{V}_{rot} + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$\vec{a}_{rot} = \vec{a}_{in} - \left(\frac{d\vec{\omega}}{dt}\right)_{rot} \times \vec{r} - 2\vec{\omega} \times \vec{V}_{rot} - \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

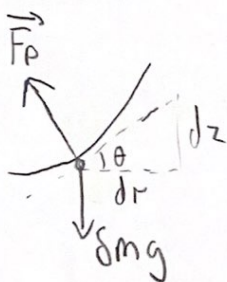
$$m\vec{a}_{rot} = \vec{F} - \underbrace{m \frac{d\vec{\omega}}{dt} \times \vec{r}}_{\text{force due to linear acceleration}} - \underbrace{2m\vec{\omega} \times \vec{V}_{rot}}_{\text{Coriolis Force}} - \underbrace{m\vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{Centrifugal force}}$$

$\downarrow$
 reel forces on the particle

* Note that we have omitted the index (rot/in) on $\frac{d\vec{\omega}}{dt}$ since it's same in both reference frames $\left(\frac{d\vec{\omega}}{dt}\right)_{in} = \left(\frac{d\vec{\omega}}{dt}\right)_{rot} + \vec{\omega} \times \vec{\omega} = \left(\frac{d\vec{\omega}}{dt}\right)_{rot}$

A diagram of a parabolic structure, possibly a liquid meniscus or a parabolic dish. The vertical axis is labeled w at the top with an upward arrow. A horizontal force F_p is applied to the right at a point on the curve. A vertical force Mg is applied downwards at the same point. The vertical distance from the bottom of the parabola to the point of application of F_p and Mg is labeled z_0 with a downward arrow.

We know that pressure gradient must be perpendicular to the surface of the liquid. Because if it had a parallel component, it would displace the liquid molecules until surface reaches an equilibrium.


$$\vec{F}_{\text{net}} = \vec{F}_p + (\delta m \vec{g}) = (F_p \cos \theta - \delta m g) \hat{z} - F_p \sin \theta \hat{r}$$

$$\vec{F}_{\text{net}} = \delta m \cdot \vec{a}_{\text{centripetal}} = \delta m \omega^2 r (-\hat{r}) \Rightarrow \text{water molecules are accelerating in the inertial RF!}$$

Then $F_p \cos \theta = \delta mg$ and $\delta m \omega^2 r = F_p \sin \theta$

$$\tan \theta = \frac{dz}{dr} = \frac{w^2 r}{g} \Rightarrow \int_{z_0}^z dz = \frac{w^2}{g} \int_0^r r dr \Rightarrow z = z_0 + \frac{w^2 r^2}{2g}$$

b) If we switch to a reference frame rotating with $\vec{\omega}$:

$$m \vec{a}_{\text{rot}} = \vec{F} - 2m \vec{\omega} \times \vec{V}_{\text{rot}} - m \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

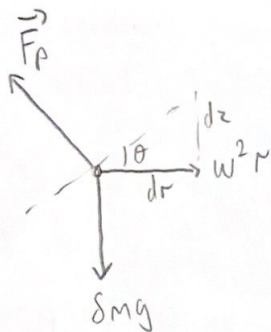
↳ In this reference frame a liquid molecule at (r, θ, z) is at rest. That means $\vec{v}_{\text{rot}} = 0$ and $\vec{a}_{\text{rot}} = 0$. (In static equilibrium)

[We ignore permutations and any other complex behaviour of the liquid]
[and assume every particle just rotates with $\vec{\omega}$]

* $\vec{F}$ is the total of real forces acting on the particle with mass δm

$$\Rightarrow \vec{F} = \delta m \vec{g} + \vec{F}_p$$

$\downarrow$
pressure gradient



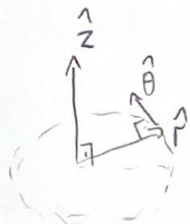
since $\vec{V}_{rot} = 0$, $\vec{a}_{rot} = 0$

$$0 = \vec{F}_p + \delta m \vec{g} - \delta m \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

→ Take $\vec{r}$ to be radial vector starting from the axis of the cylinder.

$$\vec{\omega} \times \vec{r} = \omega r \hat{\theta} ; \vec{\omega} \times (\omega r \hat{\theta}) = \omega^2 r (-\hat{r})$$

$$0 = \vec{F}_p + \delta m g (-\hat{z}) + \omega^2 r \hat{r}$$

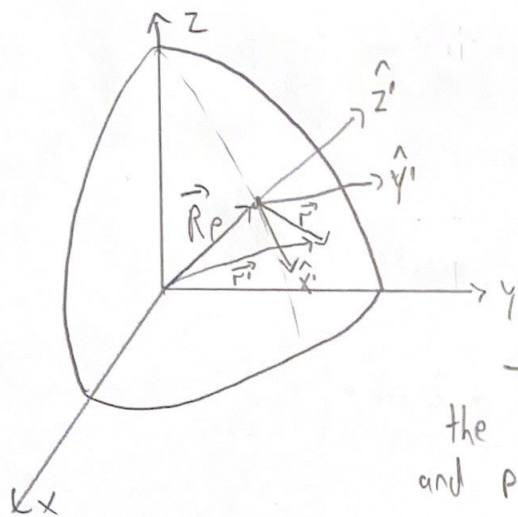


As before $F_p \cos \theta = \delta m g$ and $F_p \sin \theta = \omega^2 r$

$$\frac{dz}{dr} = \tan \theta = \frac{\omega^2 r}{g} \quad \int_{z_0}^z dz = \frac{\omega^2}{g} \int_0^r r dr$$

$$z = z_0 + \frac{\omega^2 r^2}{2g} //$$

7)



Let $\vec{r}$ be the position of any point w.r.t. center of Earth

$$\vec{r} = \vec{R}_p + \vec{r}'$$

→ At the beginning we can choose the inertial axis and rotating axis cocentric and proceed with our analysis as usual. Then we can translate the rotating axis by $\vec{R}$.

$$\left(\frac{d\vec{r}}{dt}\right)_{in} = \left(\frac{d\vec{r}}{dt}\right)_{rot} + \vec{\Omega} \times \vec{r}, \quad \left[\frac{d}{dt} \left(\frac{d\vec{r}}{dt}\right)_{in}\right]_{in} = \left[\frac{d}{dt} \left(\frac{d\vec{r}}{dt}\right)_{in}\right]_{rot} + \vec{\Omega} \times \frac{d\vec{r}}{dt}$$

$$\left(\frac{d^2\vec{r}}{dt^2}\right)_{in} = \frac{d}{dt} \left[\left(\frac{d\vec{r}}{dt}\right)_{rot} + \vec{\Omega} \times \vec{r} \right]_{rot} + \vec{\Omega} \times \left(\frac{d\vec{r}}{dt}\right)_{rot} + \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$

$$= \left[\frac{d^2\vec{r}}{dt^2}\right]_{rot} + \left[\frac{d\vec{\Omega}}{dt} \times \vec{r}\right]_{rot} + 2\vec{\Omega} \times \left(\frac{d\vec{r}}{dt}\right)_{rot} + \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$

→ $\frac{d\vec{\Omega}}{dt} = 0$ since earth rotates with const. angular velocity.

$$\left[\frac{d^2\vec{r}}{dt^2}\right]_{in} = \left[\frac{d^2\vec{r}}{dt^2}\right]_{rot} + 2\vec{\Omega} \times \left(\frac{d\vec{r}}{dt}\right)_{rot} + \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$

Now impose $\vec{r} = \vec{R} + \vec{r}' \Rightarrow \left[\frac{d^2\vec{r}}{dt^2}\right]_{in} = \left[\frac{d^2\vec{R}}{dt^2}\right]_{in} + \left[\frac{d^2\vec{r}'}{dt^2}\right]_{in}$

$\left[\frac{d^2\vec{r}'}{dt^2}\right]_{in}$ is the real forces acting on the particle $\frac{\vec{F}}{m}$

Thus $\vec{F} - \left[\frac{d^2\vec{R}_p}{dt^2}\right]_{in} = \left[\frac{d^2\vec{r}'}{dt^2}\right]_{rot} + 2\vec{\Omega} \times \left(\frac{d\vec{r}'}{dt}\right)_{rot} + \vec{\Omega} \times (\vec{\Omega} \times \vec{r}')$

$$m \vec{a}_{\text{rot}} = \left[\frac{d^2 \vec{r}}{dt^2} \right]_{\text{rot}} = \vec{F} - \left[\frac{d^2 \vec{R}_p}{dt^2} \right]_{\text{in}} - 2m \vec{\Omega} \times \left[\frac{d\vec{r}}{dt} \right]_{\text{rot}} - \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$

Now similarly for $\vec{R}$ we can write:

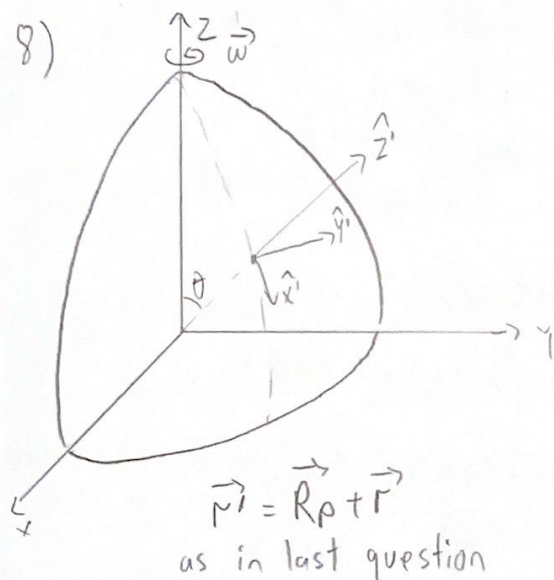
$$\left[\frac{d^2 \vec{R}_p}{dt^2} \right]_{\text{in}} = \left[\frac{d^2 \vec{R}_p}{dt^2} \right]_{\text{rot}} + \left[\frac{d\vec{\Omega}}{dt} \right]_{\text{rot}} \times \vec{R} + 2 \vec{\Omega} \times \left[\frac{d\vec{R}_p}{dt} \right]_{\text{rot}} + \vec{\Omega} \times (\vec{\Omega} \times \vec{R}_p)$$

Since $\left[\frac{d\vec{R}_p}{dt} \right]_{\text{rot}} = 0$ (an observer rotating rotating with Earth think's he's stationary)

$$\left[\frac{d^2 \vec{R}_p}{dt^2} \right]_{\text{in}} = \vec{\Omega} \times (\vec{\Omega} \times \vec{R}) \quad \text{and} \quad \left[\frac{d\vec{r}}{dt} \right]_{\text{rot}} = \vec{V}_{\text{rot}}, \quad \left[\frac{d^2 \vec{r}}{dt^2} \right]_{\text{rot}} = \vec{a}_{\text{rot}}$$

$$m \vec{a}_{\text{rot}} = \vec{F} - 2m \vec{\Omega} \times \vec{V}_{\text{rot}} - \vec{\Omega} \times (\vec{\Omega} \times \vec{R}_p) - \vec{\Omega} \times (\vec{\Omega} \times \vec{r}) //$$

8)



$$m \vec{a}_{\text{rot}} = \vec{F} - 2 \vec{w} \times \vec{v}_{\text{rot}} - w \times (\vec{w} \times \vec{r}')$$

$$\vec{F} = -G \frac{Mm}{r^2} \hat{z}$$

$$\text{Also } w \times (\vec{w} \times \vec{r}') = w \times (\vec{w} \times \vec{R}) + w \times (\vec{w} \times \vec{r}')$$

We can define g as:

$$m \vec{g} = -\frac{GMm}{R^2} \hat{z} - m \vec{w} \times (\vec{w} \times \vec{R})$$

$$\hookrightarrow \vec{r}' \approx \vec{R}_p \text{ since } R_p \gg r$$

$\rightarrow$ Also since $r \ll R$ (in areas close to Earth), and also since $|w|$ is small; we can approximate $\vec{w} \times (\vec{w} \times \vec{r}') \approx 0$

We're left with $\vec{a}_{\text{rot}} = -g \hat{z}' - 2 \vec{w} \times \vec{v}_{\text{rot}} \quad (*)$

We have $\vec{r}' = x \hat{x}' + y \hat{y}' + z \hat{z}'$

$$\vec{v}_{\text{rot}} = \frac{d\vec{r}'}{dt} = \dot{x} \hat{x}' + \dot{y} \hat{y}' + \dot{z} \hat{z}'$$

$$\vec{a}_{\text{rot}} = \frac{d^2\vec{r}'}{dt^2} = \ddot{x} \hat{x}' + \ddot{y} \hat{y}' + \ddot{z} \hat{z}'$$

$$\vec{w} \times \vec{v}_{\text{rot}} = (w \cos \theta \hat{z}' - w \sin \theta \hat{x}') \times (\dot{x} \hat{x}' + \dot{y} \hat{y}' + \dot{z} \hat{z}')$$

$$= \begin{vmatrix} \hat{x}' & \hat{y}' & \hat{z}' \\ -w \sin \theta & 0 & w \cos \theta \\ \dot{x} & \dot{y} & \dot{z} \end{vmatrix} = (-\dot{y} w \cos \theta) \hat{x}' + (\dot{x} w \cos \theta + \dot{z} w \sin \theta) \hat{y}' + (-w \dot{z} \sin \theta) \hat{z}'$$

Putting it into (*)
one gets:

$$\ddot{x} = 2 \dot{y} w \cos \theta$$

$$\ddot{y} = -2w(\dot{z} \sin \theta + \dot{x} \cos \theta)$$

$$\ddot{z} = 2w \dot{y} \sin \theta - g$$

We also have the initial conditions:

$$\begin{array}{lll} x(0)=0 & \dot{x}(0)=0 & \\ y(0)=0 & \dot{y}(0)=0 & \text{with } \ddot{x} = 2\dot{y}w\cos\theta \\ z(0)=0 & \dot{z}(0)=V_0 & \ddot{y} = -2w(\dot{x}\cos\theta + \dot{z}\sin\theta) \\ & & \ddot{z} = 2w\dot{y}\sin\theta - g \end{array}$$

Integrating once using initial conditions:

$$\left. \begin{array}{l} \dot{x} = 2yw\cos\theta \quad (1) \\ \dot{y} = -2w(x\cos\theta + z\sin\theta) \quad (2) \\ \dot{z} = 2wy\sin\theta - gt + V_0 \quad (3) \end{array} \right\} \begin{array}{l} \ddot{y} = -2w(\dot{x}\cos\theta + \dot{z}\sin\theta) \\ = -2w(2yw\cos^2\theta + 2yw\sin^2\theta + (V_0 - gt)\sin\theta) \end{array}$$

$$\ddot{y} + 4w^2y = -2w\sin\theta(V_0 - gt)$$

↳ This DE. has a homogeneous sol of the form.

$$y_h(t) = A\sin 2wt + B\cos 2wt$$

↳ particular sol, of the form $y_p(t) = \frac{-\sin\theta(V_0 - gt)}{2w}$

$$y(t) = \frac{-\sin\theta(V_0 - gt)}{2w} + A\sin(2wt) + B\cos(2wt)$$

$$y(0) = 0 \Rightarrow B = \frac{V_0\sin\theta}{2w}, \quad \dot{y}(0) = 0 \Rightarrow \frac{g\sin\theta}{2w} + A2w = 0$$

$$A = \frac{-g\sin\theta}{4w^2}$$

$$y(t) = \frac{-\sin\theta(V_0 - gt)}{2w} - \frac{g\sin\theta}{4w^2}\sin(2wt) + \frac{V_0\sin\theta}{2w}\cos(2wt)$$

put in (3) $\downarrow$

$$z(t) = \int_0^t \left[-\sin^2\theta(V_0 - gt) - \frac{g\sin^2\theta}{2w}\sin(2wt) + V_0\sin^2\theta\cos(2wt) + (V_0 - gt) \right] dt$$

$$= \left(V_0t - \frac{gt^2}{2} \right) \cos^2\theta + \frac{g\sin^2\theta}{4w^2} [\cos(2wt) - 1] + \frac{V_0\sin^2\theta}{2w}\sin(2wt)$$

$$y(t) = \frac{-\sin\theta(V_0 - gt)}{2\omega} - \frac{g\sin\theta}{4\omega^2} \sin(2\omega t) + \frac{V_0\sin\theta}{2\omega} \cos(2\omega t)$$

since $|\omega t| \ll 1$, we can Taylor expand $y(t)$

$$y(t) = \frac{-\sin\theta(V_0 - gt)}{2\omega} - \frac{g\sin\theta}{4\omega^2} \left(2\omega t - \frac{4\omega^3 t^3}{3} + \dots \right) + \frac{V_0\sin\theta}{2\omega} \left(1 - 2\omega^2 t^2 + \dots \right)$$

$$\approx \frac{-V_0\sin\theta}{2\omega} + \frac{gt\sin\theta}{2\omega} - \frac{gt\sin\theta}{2\omega} + \frac{V_0\sin\theta}{2\omega} + \frac{\omega g t^3 \sin\theta}{3} - V_0 \omega \sin\theta t^2$$

$$y(t) \approx \frac{\omega g t^3 \sin\theta}{3} - V_0 \omega \sin\theta t^2$$

$$z(t) \approx \left(V_0 t - \frac{gt^2}{2} \right) \cos^2\theta \quad \left\{ \begin{array}{l} \text{neglected } \omega t^2 \text{ and higher order terms} \\ \text{by approximating } \cos(2\omega t) \approx 1, \sin(2\omega t) \approx 0 \end{array} \right\}$$

Time elapsed till ball hits the ground $\Rightarrow z(T) = 0$

$$V_0 T - \frac{gT^2}{2} = 0 \Rightarrow T = \frac{2V_0}{g} \Rightarrow \text{what we'd expect if Earth was not rotating at all}$$

$$y(T) = \frac{\omega g \sin\theta}{3} \frac{8V_0^3}{g^3} - V_0 \omega \sin\theta \frac{4V_0^2}{g^2} = -\frac{4V_0^3 \omega \sin\theta}{3g^2} \Rightarrow \text{(-) sign means ball is deflected to the west.}$$

$$|y(T)| = \frac{4\omega}{3g^2} V_0^3 \sin\theta \text{ to the West}$$

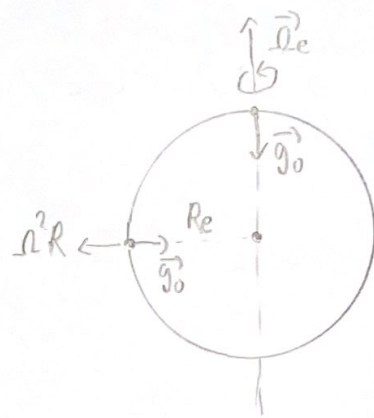
9. Apparent Force of Gravity

$$\vec{a}_{\text{rot}} = \vec{a}_{\text{in}} - 2\vec{\Omega} \times \vec{v}_{\text{rot}} - \vec{\Omega} \times (\vec{\Omega} \times \vec{r})$$

Apparently $\vec{v}_{\text{rot}} = 0$ in either case.

→ Also in poles $\vec{\Omega} \parallel \vec{r}$ so, there is no contribution from centrifugal force

→ In equator, a rotating observer would feel a centrifugal force radially outward from the axis of rotation on top of gravitational force



$$\vec{g}_{\text{pole}} = -\gamma \frac{M}{R_e^2} \hat{r}$$

$$\vec{g}_{\text{equator}} = -\gamma \frac{M}{R_e^2} \hat{r} + \Omega_e^2 R_e \hat{r}$$

$$\Delta g = g_{\text{equator}} - g_{\text{pole}} = \Omega_e^2 R_e$$

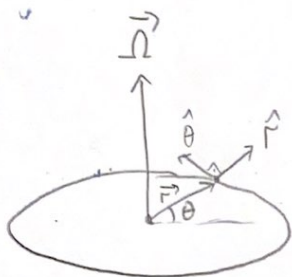
$$\Omega_e = \frac{2\pi \text{ rad}}{24 \times 3600 \text{ s}} = 7.27 \times 10^{-5} \frac{\text{rad}}{\text{s}}, \quad R_e = 6.37 \times 10^6 \text{ m}$$

$$\Delta g = \Omega_e^2 R_e = \left(7.25 \times 10^{-5} \frac{\text{rad}}{\text{s}}\right)^2 \times 6.37 \times 10^6 \text{ m} = 3.37 \times 10^{-2} \frac{\text{m}}{\text{s}^2}$$

$$\frac{\Delta g}{g} = \frac{3.37 \times 10^{-2}}{9.81} = 3.43 \times 10^{-3}$$

⇒ change in apparent gravity is very small so it's safe to neglect the centrifugal component in most cases.

10. Velocity in Polar Coordinates



Assume we are on a non-inertial rotating frame rotating with the particle.

$$\text{Then } \vec{\Omega} = \dot{\theta} \hat{k}.$$

In the rotating frame particle has only a radial velocity:

$$\vec{V}_{\text{rot}} = \dot{r} \hat{r}$$

→ It can be shown that for any vector $\vec{C}$:

$$\left(\frac{d\vec{C}}{dt} \right)_{\text{inertial}} = \left(\frac{d\vec{C}}{dt} \right)_{\text{rotational}} + \vec{\Omega} \times \vec{C}$$

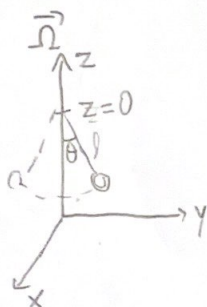
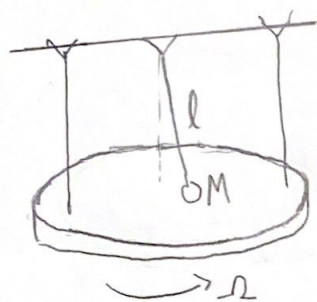
$$\text{Letting } \vec{C} = \vec{r}: \quad \left(\frac{d\vec{r}}{dt} \right)_{\text{in}} = \left(\frac{d\vec{r}}{dt} \right)_{\text{rot}} + \vec{\Omega} \times \vec{r}$$

$$\vec{V}_{\text{in}} = \vec{V}_{\text{rot}} + \vec{\Omega} \times \vec{r}$$

$$= \dot{r} \hat{r} + (\dot{\theta} \hat{k}) \times (r \hat{r}) = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}$$

$$\boxed{\vec{V}_{\text{in}} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}}$$

11) Pendulum On Rotating Platform



* In the rotating frame with basis x, y, z assume ball is oscillating only in yz plane because it's motion is constrained by a rigid rod.

$$\vec{V}_x = 0$$

→ Also set the top of the rod as $z=0$

$$\begin{aligned} * \vec{V}_{rot} &= -\dot{\theta} l \cos\theta \hat{y} + \dot{\theta} l \sin\theta \hat{z} \\ \vec{r} &= +l \sin\theta \hat{y} - l \cos\theta \hat{z} \end{aligned}$$

→ We know from our previous discussion:

$$\begin{aligned} M \left(\frac{d^2 \vec{r}}{dt^2} \right)_{rot} &= \vec{F} - 2M \vec{\Omega} \times \vec{V}_{rot} - M \vec{\Omega} \times (\vec{\Omega} \times \vec{r}) \\ &= -Mg \hat{z} - 2M \Omega \dot{\theta} l \left[\hat{z} \times (-\cos\theta \hat{y} + \sin\theta \hat{z}) \right] - M \Omega^2 l \hat{z} \times \left[\hat{z} \times (\sin\theta \hat{y} - \cos\theta \hat{z}) \right] \end{aligned}$$

$$M \left(\frac{d^2 \vec{r}}{dt^2} \right)_{rot} = -Mg \hat{z} - \underbrace{2M \Omega \dot{\theta} l \cos\theta \hat{x}}_{\text{no effect on the motion}} + M \Omega^2 l \sin\theta \hat{y}$$

no effect on the motion
because ball is constrained to move only in yz plane

→ Problem is reduced to the motion of an oscillating pendulum under the effect of gravity and centrifugal force.

$$\vec{\tau} = -Mgl \sin\theta + M \Omega^2 l^2 \sin\theta \cos\theta = I \ddot{\theta} = Ml^2 \ddot{\theta}$$

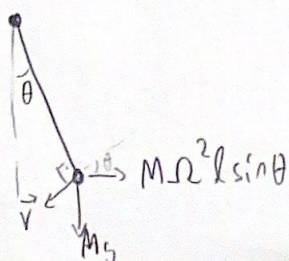
For small θ
 $\sin\theta \approx \theta, \cos\theta \approx 1$

$$-Ml(g - \Omega^2 l)\theta = Ml^2 \ddot{\theta}$$

$$\ddot{\theta} + \left(\frac{g}{l} - \Omega^2 \right) \theta = 0$$

→ oscillatory motion

$$\omega = \sqrt{\frac{g}{l} - \Omega^2}$$



* However this result is true only when $\frac{g}{l} > \Omega^2$ because it

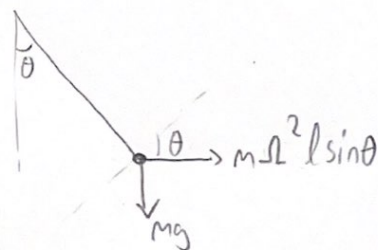
$\ddot{\theta} + \left(\frac{g}{l} - \Omega^2\right)\theta = 0$ and $\Omega^2 > \frac{g}{l}$ then this will not be an harmonic motion but an exponential motion. Without the small angle approximation:

$$\ddot{\theta} + \left(\frac{g}{l} - \Omega^2 \cos \theta\right) \sin \theta = 0$$

And assume $\Omega^2 > \frac{g}{l}$:

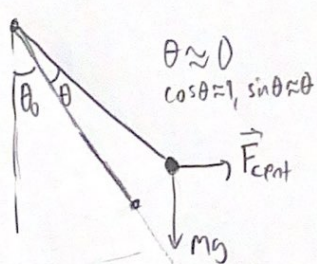
→ Then $\theta=0$ will be an unstable point and ball will accelerate towards a new equilibrium point.

Observe that ball can't go further than $\theta = \frac{\pi}{2}$ because torque of the centrifugal force decreases as θ increases.



At this new equilibrium $\Rightarrow Mg = M\Omega^2 l \cos \theta_0$

$\theta_0 = \cos^{-1}\left(\frac{g}{\Omega^2 l}\right)$ is the new equilibrium position.



$\theta \approx 0$
 $\cos \theta \approx 1, \sin \theta \approx \theta$

$$\tau = -Mgl \sin(\theta_0 + \theta) + M\Omega^2 l^2 \sin(\theta_0 + \theta) \cos(\theta_0 + \theta) = Ml^2 \ddot{\theta}$$

$$\Rightarrow (\sin \theta \cos \theta_0 + \sin \theta_0 \cos \theta) \left[-\Omega^2 (\cos \theta \cos \theta_0 - \sin \theta \sin \theta_0) - \frac{g}{l} \right] = \ddot{\theta}$$

$$\left[-\Omega^2 \frac{g}{\Omega^2 l} - \Omega^2 \theta \sin \theta_0 - \frac{g}{l} \right] \left(\theta \frac{g}{\Omega^2 l} + \sin \theta_0 \right) = \ddot{\theta}$$

can be neglected
order 2

$$-\theta^2 \sin \theta_0 \frac{g}{l} - \Omega^2 \theta \sin^2 \theta_0 = \ddot{\theta} \Rightarrow \ddot{\theta} + (\Omega^2 \sin^2 \theta_0) \theta = 0$$

$$\omega'^2 = \Omega^2 (1 - \cos^2 \theta_0) = \Omega^2 \left(1 - \frac{g^2}{\Omega^4 l^2}\right) = \Omega^2 - \frac{g^2}{\Omega^2 l^2} \Rightarrow \omega' = \sqrt{\Omega^2 - \frac{g^2}{\Omega^2 l^2}}$$